COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S4
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

Progress Report

main.cpp

main.cpp > ...

```cpp
#include <iostream>
#include "bst.h"

int main() {
    //create the root node:
    BinarySearchTree<int> tree(20);

    tree.insert(10);
    tree.insert(5);
    tree.insert(70);
    tree.insert(80);
    tree.insert(40);
    tree.insert(50);

    std::cout << "preOrder: ";
    tree.preOrder();
    std::cout << std::endl;

    std::cout << "inOrder: ";
    tree.inOrder();
    std::cout << std::endl;

    std::cout << "postOrder: ";
    tree.postOrder();
    std::cout << std::endl;

    return 0;
}
```

BST.h

bst.h > BinarySearchTree(T)

```h
#ifndef BST_H
#define BST_H

template <typename T>
class BinarySearchTree{
    private:
    T key;
    // pointers:
    BinarySearchTree<T>* left;
    BinarySearchTree<T>* right;

    public:
    //constructor:
    BinarySearchTree(T value);
    void insert(T value);

    //traversing algo:
    void preOrder();
    void inOrder();
    void postOrder();

};
```

```
23
24  template <typename T>
25  BinarySearchTree<T>::BinarySearchTree(T value){
26      key = value;
27      left = nullptr;
28      right = nullptr;
29  }
30
31  template <typename T>
32  void BinarySearchTree<T>::insert(T value){
33      if(value < key){
34          //do something
35          if(left == nullptr){
36             left = new BinarySearchTree<T>(value);
37          } else {
38              left->insert(value);
39          }
40      } else if (value > key) {
41          // do something
42          if(right == nullptr){
43              right = new BinarySearchTree<T>(value);
44          } else {
45              right->insert(value);
46          }
47      }
48  }
49
50  template <typename T>
51  void BinarySearchTree<T>::preOrder(){
52      std::cout << key << " ";
53      if(left != nullptr){
54          left->preOrder();
55      }
56      if(right != nullptr){
57          right->preOrder();
58      }
59  }
```

```
60
61  template <typename T>
62  void BinarySearchTree<T>::inOrder(){
63
64      //go to left
65      if(left != nullptr){
66          left->inOrder();
67      }
68      std::cout<<key<< " ";
69      //go to right
70      if(right != nullptr){
71          right->inOrder();
72      }
73  }
74
75  template <typename T>
76  void BinarySearchTree<T>::postOrder(){
77
78      //go to left
79      if(left != nullptr){
80          left->postOrder();
81      }
82      //go to right
83      if(right != nullptr){
84          right->postOrder();
85      }
86      std::cout<<key<< " ";
87  }
88
89  #endif
```

Output

```
preOrder: 20 10 5 70 40 50 80
inOrder: 5 10 20 40 50 70 80
postOrder: 5 10 50 40 80 70 20
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S4-UCHI\HOA8>
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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